

DNA Structure: Pair, Twist, Critique

I can construct a complementary DNA section and describe the strengths and limitations of the model.

PAIR

A-T and C-G

TWIST

double helix

CRITIQUE

useful but simplified

[Open the original interactive lesson](#)

What success looks like

- I can state that DNA has two strands coiled into a double helix.
- I can apply A-T and C-G complementary base pairing.
- I can identify at least one model strength and limitation.

Retrieve and reconnect

- Describe the chromosome-DNA-gene relationship.
- Define genome.
- State one limit of the hierarchy model.
- Lead-in: predict the partner for base C.

One DNA strand reads A-C-G-T. Which complementary strand fits the pairing rule?

Think first. Point, say, sign or write your choice. Give one reason.

- A. T-G-C-A
- B. A-C-G-T
- C. A-A-A-A

Look closely · what is the evidence?

A pairs with T

C pairs with G

Two paired strands coil into a double helix

Look closely · what is the evidence?



A T pairs



C G pairs



Ladder

Words we will use

- nucleotide — one DNA building unit
- complementary — fits by a pairing rule
- double helix — two-strand twist

Watch the model

- DNA is a polymer of nucleotides. Each nucleotide contains a sugar, a phosphate group and one base; nucleotides join to form each sugar-phosphate backbone.
- Build the second strand using complementary base pairs: adenine (A) pairs with thymine (T), and cytosine (C) with guanine (G); weak hydrogen bonds join each pair between the strands.
- Twist the paired ladder gently to represent the double helix while keeping the base order visible.

A short strand reads G-A-T-C. Which complementary sequence is correct?

- A. C-T-A-G
- B. G-A-T-C
- C. A-C-T-G

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Strength: the ladder shows two strands, base order and complementary pairing clearly.
- Limit: colours, distances and component sizes are invented; a classroom twist is not the molecule's true scale or movement.
- A gene is a particular sequence along DNA; do not add Higher-tier transcription or translation stages to this Foundation lesson.

Find and repair the misconception

Complementary DNA strands have exactly the same base sequence.

- A. Keep it: complementary means identical.
- B. Repair it: each base fixes its partner, A-T and C-G.
- C. Repair it: every base can pair with any other base.

Complementary base-pair lab

Match each base to its complementary partner, freeze the sequence, then critique one feature of the model.

- READ
- PAIR
- CHECK
- TWIST
- CRITIQUE

Complementary base-pair lab

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- A ↔ T PAIR
- C ↔ G PAIR
- MODEL LIMIT

Read the two sequences, then name what the model helps us see and what it cannot show.

Your independent mission

Construct a complementary DNA section, annotate its structure and evaluate the model.

Record: A correct complementary sequence, labelled two-strand/double-helix description and one valid model limitation.

Choose your support and challenge

- Supported: Complete a six-base sequence with A-T and C-G cards, then point to the two strands.
- Standard: Complete a ten-base complementary sequence and describe nucleotide, two strands and double helix.
- Stretch: Construct and check the sequence, then evaluate two useful features and two limitations of the model.

Show what you know

Write the complement of A-G-C-T.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.